

Robust Linear Regression: Optimal Rates in Polynomial Time

Ainesh Bakshi

abakshi@cs.cmu.edu
Carnegie Mellon University
Pittsburgh, USA

Adarsh Prasad

adarshp@andrew.cmu.edu
Carnegie Mellon University
Pittsburgh, USA

ABSTRACT

We obtain robust and computationally efficient estimators for learning several linear models that achieve statistically optimal convergence rates under minimal distributional assumptions. Concretely, we assume our data is drawn from a k -hypercontractive distribution and an ϵ -fraction is adversarially corrupted. We then describe an estimator that converges to the optimal least-squares minimizer for the true distribution at a rate proportional to $\epsilon^{2-2/k}$, when the noise is independent of the covariates. We note that no such estimator was known prior to our work, even with access to unbounded computation. The rate we achieve is information-theoretically optimal, and thus we resolve the main open question in Klivans, Kothari, and Meka [COLT'18].

Our key insight is to identify an analytic condition that serves as a polynomial relaxation of independence of random variables. In particular, we show that when the moments of the noise and covariates are negatively correlated, we obtain the same rate as independent noise. Further, when the condition is not satisfied, we obtain a rate proportional to $\epsilon^{2-4/k}$, and again match the information-theoretic lower bound. Our central technical contribution is to algorithmically exploit independence of random variables in the “sum-of-squares” framework by formulating it as the aforementioned polynomial inequality.

CCS CONCEPT

• Theory of computation → Semidefinite programming; Convex optimization; • Mathematics of computing → Multivariate statistics.

KEYWORDS

robust linear regression, heavy-tailed distributions, sum-of-squares

Acm Reference Format

Ainesh Bakshi and Adarsh Prasad. 2021. Robust Linear Regression: Optimal Rates in Polynomial Time. In Proceedings of the 53rd Annual ACM SIGACT Symposium on Theory of Computing (STOC '21), June 21–25, 2021, Virtual, Italy. ACM, New York, NY, USA, 14 pages. <https://doi.org/10.1145/3406325>



This work is licensed under a Creative Commons Attribution International 4.0 License.

STOC '21, June 21–25, 2021, Virtual, Italy

© 2021 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-8053-9/21/06.

<https://doi.org/10.1145/3406325>

1 INTRODUCTION

This paper addresses the problem of robust linear regression, which involves creating regression models that remain accurate even when some data is corrupted by adversarial outliers. Traditional regression methods fail under these conditions, making robust estimation a challenging task. The authors focus on a hypercontractive distribution framework, which includes distributions like Gaussian, allowing them to design robust estimators. Their estimators achieve optimal convergence rates (best possible accuracy) even with corrupted data, something previous methods couldn't do. Additionally, the estimators are computationally efficient, working in polynomial time using the sum-of-squares convex hierarchy. They also show that relying on just bounded covariance assumptions is insufficient for robust regression in adversarial settings. Their approach balances robustness, accuracy, and computational complexity, making it suitable for high-dimensional data. Prior methods, while statistically sound, were often computationally intractable in these scenarios. The paper's key contribution is a unified framework that handles various noise models and outperforms previous estimators. Ultimately, the work provides a significant advancement in robust statistics by achieving both theoretical and practical efficiency.

1.1 Our Results

1.1.1 Model 1.1: Robust Regression Model

Let $\mathcal{D}$ be a distribution over $\mathbb{R}^d \times \mathbb{R}$ such that the marginal distribution over $\mathbb{R}^d$ is centered and has covariance Σ^* , and let

$$\Theta^* = \arg \min_{\Theta} \mathbb{E}_{(x,y) \sim \mathcal{D}} [(y - \langle \Theta, x \rangle)^2]$$

be the optimal hyperplane for $\mathcal{D}$. Let $\{(x_1^*, y_1^*), (x_2^*, y_2^*), \dots, (x_n^*, y_n^*)\}$ be n i.i.d. random variables drawn from $\mathcal{D}$. Given $\epsilon > 0$, the robust regression model $\mathcal{R}_{\mathcal{D}}(\epsilon, \Sigma^*, \Theta^*)$ outputs a set of n samples $\{(x_i, y_i)\}_{i \in [n]}$ such that for at least $(1 - \epsilon)n$ points, $x_i = x_i^*$ and $y_i = y_i^*$. The remaining ϵn points are arbitrary and potentially adversarial with respect to the input and estimator.

1.1.2 Theorem 1.2: Bounded Covariance does not suffice (Theorem 6.3 informal)

For all $\epsilon > 0$, there exist two distributions $\mathcal{D}_1, \mathcal{D}_2$ over $\mathbb{R}^d \times \mathbb{R}$ such that

$$d_{TV}(\mathcal{D}_1, \mathcal{D}_2) \leq \epsilon$$

and the marginal distribution over the covariates has bounded covariance, denoted by $\Sigma_2 = \Theta(1)$, yet

$$\Sigma^{1/2}(\Theta_1 - \Theta_2)^2 = \Omega(1),$$

where Θ_1 and Θ_2 are the optimal hyperplanes for $\mathcal{D}_1$ and $\mathcal{D}_2$, respectively.

Definition 1.3 ((C, k) -Hypercontractivity) A distribution D over $\{-1, 1\}^n$ is (C, k) -hypercontractive for an even integer $k \geq 2$, [...]

$$r \in [k/2], \text{ for all } x \in \mathbb{R}^d, \mathbb{E}_{x \sim D} \left[\left\langle v, x - \mathbb{E}[x] \right\rangle^2 \right] \leq r^2 \mathbb{E}_{x \sim D} \left[\left\langle c_x - \mathbb{E}[x] \right\rangle^2 \right]^r \quad (1)$$

Theorem 1.7 (Robust Regression with Negatively Correlated Noise, Theorem 5.1 informal). Given $\epsilon > 0$, $k \geq 4$, and $n \geq (d \log(d))^{O(k)}$ samples from $\mathcal{R}_D(\epsilon, \Sigma^*, \Theta^*)$, such that $\mathcal{D}$ is (c, k) -certifiably hypercontractive and has negatively correlated moments, there exists an algorithm that runs in $n^{O(k)}$ time and outputs an estimator $\tilde{\Theta}$ such that with high probability,

$$(\Sigma^*)^{1/2} \|\Theta^* - \tilde{\Theta}\|_2 \leq O\left(\epsilon^{1-1/k}\right) \text{err}_D(\Theta^*)^{1/2}.$$

and ,

$$\text{err}_D(\tilde{\Theta}) \leq \left(1 + O\left(\epsilon^{2-2/k}\right)\right) \text{err}_D(\Theta^*).$$

Theorem 1.9 (Lower Bound for Independent Noise, Theorem 6.1 informal). For any $\epsilon > 0$, there exist two distributions $\mathcal{D}_1, \mathcal{D}_2$ over $\mathbb{R}^2 \times \mathbb{R}$ such that the marginal distribution over $\mathbb{R}^2$ has covariance Σ and is (c, k) -hypercontractive for both distributions, and yet

$$(\Sigma)^{1/2} \|\Theta_1 - \Theta_2\|_2^2 = \Omega\left(\epsilon^{1-1/k} \sigma\right),$$

where Θ_1 and Θ_2 are the optimal hyperplanes for $\mathcal{D}_1$ and $\mathcal{D}_2$, respectively, $\sigma = \max(\text{err}_{\mathcal{D}_1}(\Theta_1), \text{err}_{\mathcal{D}_2}(\Theta_2))$, and the noise is uniform over $[-\sigma, \sigma]$. Further,

$$|\text{err}_{\mathcal{D}_1}(\Theta_2) - \text{err}_{\mathcal{D}_1}(\Theta_1)| = \Omega\left(\epsilon^{2-2/k} \sigma^2\right).$$

Corollary 1.10 (Robust Regression with Dependent Noise, Corollary 4.2 informal). Given $\epsilon > 0$, $k \geq 4$, and $n \geq d^{O(k)}$ samples from $\mathcal{R}_D(\epsilon, \Sigma^*, \Theta^*)$, such that $\mathcal{D}$ is (c, k) -certifiably hypercontractive, there exists an algorithm that runs in $n^{O(k)}$ time and outputs an estimator $\tilde{\Theta}$ such that with probability $9/10$,

$$(\Sigma^*)^{1/2} \|\Theta^* - \tilde{\Theta}\|_2 \leq O\left(\epsilon^{1-2/k}\right) \text{err}_D(\Theta^*)^{1/2},$$

and

$$\text{err}_D(\tilde{\Theta}) \leq \left(1 + O\left(\epsilon^{2-4/k}\right)\right) \text{err}_D(\Theta^*).$$

Further, we show that the dependence on ϵ is again information-theoretically optimal:

Theorem 1.11 (Lower Bound for Dependent Noise, Theorem 6.2 informal). For any $\epsilon > 0$, there exist two distributions $\mathcal{D}_1, \mathcal{D}_2$ over $\mathbb{R}^2 \times \mathbb{R}$ such that the marginal distribution over $\mathbb{R}^2$ has covariance Σ and is (c, k) -hypercontractive for both distributions, and yet

$$(\Sigma)^{1/2} \|\Theta_1 - \Theta_2\|_2^2 = \Omega\left(\epsilon^{1-2/k} \sigma\right),$$

where Θ_1 and Θ_2 are the optimal hyperplanes for $\mathcal{D}_1$ and $\mathcal{D}_2$, respectively, and $\sigma = \max(\text{err}_{\mathcal{D}_1}(\Theta_1), \text{err}_{\mathcal{D}_2}(\Theta_2))$. Further,

$$|\text{err}_{\mathcal{D}_1}(\Theta_2) - \text{err}_{\mathcal{D}_1}(\Theta_1)| = \Omega\left(\epsilon^{2-4/k} \sigma^2\right).$$

Corollary 1.12 (Robust Regression with Gaussian Covariates). Given $\epsilon > 0$ and $n \geq (d \log(d))^{O(\log(1/\epsilon))}$ samples from $\mathcal{R}_{\mathcal{N}}(\epsilon, \Sigma^*, \Theta^*)$, such that the marginal distribution over the x 's is $\mathcal{N}(0, \Sigma^*)$, there exists an algorithm that runs in $n^{O(\log(1/\epsilon))}$ time and outputs an estimator $\tilde{\Theta}$ such that with high probability,

$$(\Sigma^*)^{1/2} \|\Theta^* - \tilde{\Theta}\|_2 \leq O(\epsilon \log(1/\epsilon)) \text{err}_{\mathcal{N}}(\Theta^*)^{1/2},$$

and

$$\text{err}_{\mathcal{N}}(\tilde{\Theta}) \leq \left(1 + O\left((\epsilon \log(1/\epsilon))^2\right)\right) \text{err}_{\mathcal{N}}(\Theta^*).$$

We note that our estimators obtain the rate matching recent work for Gaussians, albeit in quasi-polynomial time. In comparison, Diakonikolas, Kong, and Stewart [21] obtain the same rate in polynomial time, when the noise is independent of the covariates. We note that obtaining the optimal rate for Gaussian covariates (shaving the additional $\log(1/\epsilon)$ factor) remains an outstanding open question.

Table 1: Comparison of convergence rate (for least-squares error) achieved by various computationally efficient estimators for Robust Regression, when the underlying distribution is $\mathcal{C}_{\mathcal{K}}, \mathcal{K}$ -hypercontractive

Estimator	Independent Noise	Arbitrary Noise
Prasad et al.[7] Diakonikolas et. al.[10]	$\epsilon \kappa^2$ (only $k=4$)	$\epsilon \kappa^2$ (only $k=4$)
Klivans, Kothari, and Meka [5]	$\epsilon^{1-2/k}$	$\epsilon^{1-2/k}$
Zhu,Jiao and Steinhardt [6]	$\epsilon^{2-4/k}$	$\epsilon^{2-4/k}$
Our Work Thm 1.7, Cor 1.10	$\epsilon^{2-2/k}$	$\epsilon^{2-4/k}$
Lower Bounds Thm 1.9, Thm 1.11	$\epsilon^{2-2/k}$	$\epsilon^{2-4/k}$

Over samples from $\mathcal{N}(0, \Sigma)$ are $(O(k), O(k))$ -certifiably hypercontractive for all k (see Section 5 in Kothari and Steurer [36]). As a consequence, instantiating Corollary 1.10 with $k = \log(1/\epsilon)$ yields the following:

1.2 Related Works

Robust Statistics. Computationally efficient estimators for robust statistics in high dimension have been extensively studied, following the initial work on robust mean estimation [[12], [3]]. We focus on literature regarding robust regression and sum-of-squares. We refer the reader to recent surveys and theses for an extensive discussion of the literature on robust statistics [[9], [1], [2], [8]].

2 Robust Certifiability and Information Theoretic Estimators

In this section, we provide an estimator that obtains the information theoretically optimal rate for robust regression. We note that we consider the setting where both the covariates and the noise are hypercontractive and are independent of each other. This setting displays all the key ideas of our estimator. Further, our estimator extends to the remaining settings, such as bounded dependent noise, by simple modifications to the subsequent analysis.

Theorem 4.1 (Robust Certifiability with Optimal Rate). Given $\epsilon > 0$, let D and D' be distributions over $\mathbb{R}^d \times \mathbb{R}$ such that the respective marginal distributions over $\mathbb{R}^d$, denoted by D_X and D'_X , are (ck, k) -hypercontractive and $\|D - D'\|_{\text{TV}} \leq \epsilon$. Let $R_D(\epsilon, \Sigma_D, \Theta_D)$ and $R_{D'}(\epsilon, \Sigma_{D'}, \Theta_{D'})$ be the corresponding instances of robust regression such that D and D' have negatively correlated moments. Further, for $(x, y) \sim D, D'$, let the marginal distribution over $y - x$, $\mathbb{E}[xx^\top]^{-1}\mathbb{E}[xy]$, be $(\eta k, k)$ -hypercontractive. Then,

$$\|\Sigma_D^{1/2}(\Theta_D - \Theta_{D'})\|_2^2 \leq O\left(\sqrt{ck\eta k\epsilon^{1-1/k}}\left(\text{err}_D(\Theta_D)^{1/2} + \text{err}_{D'}(\Theta_{D'})^{1/2}\right)\right).$$

Further,

$$\text{err}_D(\Theta_{D'}) \leq \left(1 + O\left(ck\eta k\epsilon^{2-2/k}\right)\right)\text{err}_D(\Theta_D) + O\left(ck\eta k\epsilon^{2-2/k}\right)\text{err}_{D'}(\Theta_{D'}).$$

Proof. Consider a maximal coupling of D and D' over $(x, y) \times (x', y')$, denoted by G , such that the marginal of G on (x, y) is D , the marginal on (x', y') is D' , and $P_G[I\{(x, y) = (x', y')\}] = 1 - \epsilon$. Then, for all v ,

$$\begin{aligned}\langle v, \Sigma_D(\Theta_D - \Theta_{D'}) \rangle &= \mathbb{E}_G \left[\langle v, xx^\top (\Theta_D - \Theta_{D'}) + xy - x'y' \rangle \right] \\ &= \mathbb{E}_G [\langle v, x(\langle x, \Theta_D \rangle - y) \rangle] + \mathbb{E}_G [\langle v, x(y - \langle x, \Theta_{D'} \rangle) \rangle].\end{aligned}$$

Since Θ_D is the minimizer for the least squares loss, we have the following gradient condition: for all $v \in \mathbb{R}^d$,

$$\mathbb{E}_{(x,y) \sim D} [\langle v, (\langle x, \Theta_D \rangle - y)x \rangle] = 0.$$

Since G is a coupling, using the gradient condition and that $1 = I\{(x, y) = (x', y')\} + I\{(x, y) \neq (x', y')\}$, we can rewrite equation (12) as

Algorithm: Optimal Robust Regression in Polynomial Time

Input: n samples $\mathcal{X}_\epsilon$ from the robust regression model

$\mathcal{R}_D(\epsilon, \Theta^*, \Sigma^*)$.

Operation:

1. Find a degree- $\mathcal{O}(k)$ pseudo-distribution $\tilde{\zeta}$ satisfying $\mathcal{A}_{\epsilon, \lambda_k}$ and minimizing

$$\min_{w, x', y', \Theta} \mathbb{E}_{\tilde{\zeta}} \left[\left(\frac{1}{n} \sum_{i \in [n]} w_i (y'_i - \langle \Theta, x'_i \rangle)^2 \right)^k \right].$$

2. Round the pseudo-distribution to obtain an estimator $\mathbb{E}_{\tilde{\zeta}}[\Theta]$.

Output: A vector $\mathbb{E}_{\tilde{\zeta}}[\Theta]$ such that the recovery guarantee in Theorem [11] is satisfied.

At a high level, we do not enforce the negatively correlated moments constraint in our polynomial system $\mathcal{A}_{\epsilon, \lambda_k}$ and instead use the SoS Cauchy-Schwarz inequality in our key technical lemma (Lemma ??). For completeness, we provide the proof of the SoS lemma in Appendix ??.

Lower bounds

In this section, we present information-theoretic lower bounds on the rate of convergence of parameter estimation and least-squares error for robust regression. Our constructions proceed by demonstrating two distributions over regression instances that are ϵ -close in total variation distance and the marginal distribution over the covariates is hypercontractive, yet the true hyperplanes are $f(\epsilon)$ -far in scaled ℓ_2 distance.

Agnostic Model

Next, consider the setting where we simply observe samples from $(x, y) \sim \mathcal{D}$, and our goal is to return is to return the minimizer of the squared error, given by $\Theta^* = \mathbb{E} [xx^\top]^{-1} \mathbb{E} [xy]$. Here, the distribution of the noise is allowed to depend on the covariates arbitrarily. We further assume the noise is hypercontractive and obtain a lower bound proportional to $\epsilon^{1-2/k}$ for recovering an estimator close to Θ^* . This matches the upper bound obtained in Corollary [4].

Definition 2.1 (Student's t -distribution). Given $\nu > 1$, Student's t -distribution has the following probability density function:

$$f_\nu(t) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi} \Gamma(\frac{\nu}{2})} \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

where $\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx$, for $z \in \mathbb{R}$, is the Gamma function.

References

- [1] Jason Altschuler, Victor-Emmanuel Brunel, and Alan Malek. “Best Arm Identification for Contaminated Bandits”. In: *arXiv preprint arXiv:1802.09514* (2018).
- [2] Samuel B Hopkins. “Sub-Gaussian Mean Estimation in Polynomial Time”. In: *arXiv preprint arXiv:1809.07425* (2018).
- [3] Guillaume Lecué and Jules Depersin. “Robust subgaussian estimation of a mean vector in nearly linear time”. In: *arXiv preprint arXiv:1906.03058* (2019).
- [4] Liu Liu et al. “High Dimensional Robust Sparse Regression”. In: *arXiv preprint arXiv:1805.11643* (2018).
- [5] John Ashworth Nelder and Robert WM Wedderburn. “Generalized linear models”. In: *Journal of the Royal Statistical Society: Series A (General)* 135.3 (1972), pp. 370–384.
- [6] Adarsh Prasad, Sivaraman Balakrishnan, and Pradeep Ravikumar. “A unified approach to robust mean estimation”. In: *arXiv preprint arXiv:1907.00927* (2019).
- [7] Adarsh Prasad et al. “Robust estimation via robust gradient estimation”. In: *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 82.3 (2020), pp. 601–627. DOI: [10.1111/rssb.12364](https://doi.org/10.1111/rssb.12364). eprint: <https://rss.onlinelibrary.wiley.com/doi/pdf/10.1111/rssb.12364>. URL: <https://rss.onlinelibrary.wiley.com/doi/abs/10.1111/rssb.12364>.
- [8] Roman Vershynin. “Introduction to the non-asymptotic analysis of random matrices”. In: *arXiv preprint arXiv:1011.3027* (2010).
- [9] Martin J Wainwright. *High-dimensional statistics: A non-asymptotic viewpoint*. Vol. 48. Cambridge University Press, 2019.
- [10] Sanford Weisberg. *Applied linear regression*. Vol. 528. John Wiley & Sons, 2005.

- [11] Huan Xu, Constantine Caramanis, and Shie Mannor. “Outlier-robust PCA: the high-dimensional case”. In: *IEEE transactions on information theory* 59.1 (2013), pp. 546–572.
- [12] Banghua Zhu, Jiantao Jiao, and Jacob Steinhardt. “Generalized resilience and robust statistics”. In: *arXiv preprint arXiv:1909.08755* (2019).